

MASTERCARD INNOVATION CHALLENGE 2026 | SOLUTION WALKTHROUGH

AegisLoop

Closing the loop between emerging payment-fraud campaigns and adaptive defense

Abstract

Payment-fraud defenses are usually evaluated against a fixed attack set, while adversaries adapt after each control change. AegisLoop turns that asymmetry into a controlled learning loop: it identifies eight equal-priority GenAI-enabled fraud archetypes and 32 variants; generates stateful, entity-linked synthetic campaigns behind a fidelity firewall; uses a LinUCB contextual bandit to select complete campaign parameters; and hardens a supervised, unsupervised and relational-risk defender on valid evasions. In the fixed five-seed full protocol, the bandit achieved mean reward 0.807 (95% CI 0.519-1.212), exceeding random search on every paired seed (mean difference 0.681; one-sided paired Wilcoxon $p=0.03125$). Held-out-family ROC-AUC rose from 0.9806 at V0 to 0.99993 at V2, while recall rose from 86.59% to 99.90%. Against a matched, fixed action pool held constant across generations, the hardened defender (V2) detected every campaign across five seeds. The central caveat is operationally important: V2 precision falls from 90.17% at 9.39% laboratory prevalence to 1.31% when the same scores are reweighted to 0.015% prevalence (seed 20260812; full range in Appendix Table A3). The contribution is therefore not a production detector claim; it is a reproducible campaign-level extension of single-transaction RL evasion that closes the loop into adversarial defender hardening. [E1-E6]

The decision in one sentence

Use AegisLoop as an offline, governed threat laboratory that finds campaign-level blind spots before they become production losses - with evidence boundaries made explicit.

What the evidence says	Five-seed full-protocol result	Interpretation
Adaptive search	Bandit 0.807 vs random 0.126 reward	All five paired seeds favor bandit; $p=0.03125$ [E1,E2]
Defender hardening	AUC 0.9806 -> 0.99993; recall 86.59% -> 99.90%	Held-out families improve through V2 [E1]
Honesty check	V2 precision 90.17% -> 1.31%	Seed 20260812; prevalence, not AUC, changes the operational story [E4]

How the paper maps to judging criteria

Criterion	Where it is answered	Evidence boundary
Attack diversity	Section 3 + Figure 2	Eight equal archetypes, 32 variants [E6]
Simulation fidelity	Section 5	Structural validity only; no real-data equivalence [E5,E7]
Detection efficacy	Sections 8-9	Five-seed full protocol; held-out families [E1-E4]
Novelty	Sections 4 and 6	Campaign-level RL-to-hardening loop; no unqualified 'first'
Feasibility / scale / viability	Section 10	Qualitative architecture; no unsupported latency or ROI

1 | Introduction

The material problem is not detecting yesterday's fraud; it is creating a safe, measurable loop in which new attack hypotheses expose gaps and those gaps improve the next defender.

The challenge asks teams to identify novel fraud attacks, generate high-fidelity simulations, build effective mitigations and demonstrate real-world feasibility. It also states that defense gaps should feed new attack ideas. AegisLoop treats those requirements as one system rather than four disconnected artifacts. [C1]

The three-pillar problem

- **Identify:** maintain a structured attack atlas that spans authorization, account-to-account, identity, merchant and post-transaction abuse.
- **Generate:** compile each hypothesis into an ordered, entity-linked synthetic campaign whose validity can be checked before it receives reward.
- **Defend:** let an adaptive policy probe a frozen detector, then retrain and recalibrate only at an explicit governance boundary.

Falsifiable hypotheses

The evaluation tests four explicit hypotheses under the fixed protocol; they are not claims inferred after looking at a single favorable seed. [E1,E2,E8]

- **H1 - Adaptive search:** with the defender, action pool and campaign budget held fixed, LinUCB produces higher mean campaign reward than random search and rule mutation.
- **H2 - Adversarial hardening:** training on fidelity-valid evasions improves held-out-family ranking and recall from V0 through the hardened generations without using the test split for threshold selection.
- **H3 - Later-generation stability:** a second hardening transition does not increase matched-pool fresh-attacker reward relative to V1; with five seeds, this test is directional and low-powered.
- **H4 - Novelty contribution:** zeroing the novelty bonus changes attacker reward; the two-sided test permits a positive, negative or null answer.

Citation key: [R1]-[R5] denote public references, [E1]-[E10] denote versioned project evidence, and [C1] denotes the competition rules.

Contributions

- **1.** A campaign-level attack representation: actions select a family plus intensity, timing, reuse and evasion parameters, rather than perturbing one transaction feature at a time. [E7]
- **2.** A hand-coded, stateful payment digital twin that preserves chronology and entity reuse by construction, avoiding the structural limitations of row-independent generators without claiming real-data fidelity. [E7,R3]
- **3.** A tractable contextual-bandit formulation with immediate campaign reward, equal-budget baselines and per-term reward logging. [E1,E2,E7,E8]
- **4.** A leakage-controlled V0->V1->V2 hardening protocol with final held-out families, validation-only threshold calibration and raw campaign diagnostics. [E1-E5,E8]
- **5.** A negative-result discipline: the novelty reward term is inert, the V1->V2 stability result is directional rather than conclusive, and deployment-prevalence precision collapses. [E1,E4]

2 | Related Work and Precise Positioning

AegisLoop does not claim to invent RL evasion, contextual bandits or relational fraud scoring; its defensible novelty is connecting those ideas at the campaign and defender-generation levels.

FRAUD-RLA establishes the closest adversarial-RL precedent.

Lunghi et al. formulate reinforcement-learning attacks that bypass credit-card fraud classifiers under constrained attacker knowledge. AegisLoop adopts the adversarial learning premise but changes the unit of action: the policy selects a complete, linked campaign rather than only feature-level evasion for a transaction. It then carries successful, fidelity-valid evasions across a governed boundary into defender hardening. [R1,E7]

The Adyen study supports the bandit choice and motivates the V2 stability test.

Vangara and Egg analyze contextual bandits in payment processing, including non-uniform exploration, delayed feedback and later-generation instability. Their application optimizes payment decisions; AegisLoop uses LinUCB adversarially to choose attack campaigns. The paper's warning about reward-distribution shift is why AegisLoop tests a second hardening transition instead of stopping at V1. [R2,E1,E8]

Sajja explains why the simulator is stateful and hand-coded - but does not validate it.

Sajja shows that row-independent tabular generators are structurally unable to preserve multi-account graph motifs and positive within-entity inter-event-time autocorrelation. AegisLoop therefore uses a PaySim-style agent simulation that binds entities once and evolves sequences. This is a design justification, not evidence that the synthetic campaigns match real fraud; no real reference corpus is used. [R3,E7]

Mastercard's public direction makes relational risk the right deployment interface.

Mastercard describes Decision Intelligence Pro as real-time decisioning that connects account, purchase, merchant and device relationships. AegisLoop does not claim equivalence to DI Pro and uses no published Mastercard performance number as a benchmark. The alignment is architectural: entity relationships and transaction context should enrich an issuer or network score, while AegisLoop remains an offline pre-production red-team laboratory. [R4,E7]

Reference	What it establishes	AegisLoop's distinct scope
FRAUD-RLA [R1]	RL can evade fraud classifiers	Campaign-level, entity-linked actions plus hardening
Adyen bandits [R2]	Industrial bandit trade-offs and instability	Adversarial search; explicit V1->V2 check
Sajja [R3]	Limits of row-independent synthesis	Stateful hand-coded simulation; no fidelity proof
Mastercard DI Pro [R4]	Relational, contextual risk direction	Offline discovery feeds governed score enhancement

3 | Identify | Threat Landscape

The atlas is deliberately broad and symmetric: all eight archetypes are first-class hypotheses, while three case studies simply make the mechanics concrete.

Attack atlas: equal coverage across the payment lifecycle

Eight archetypes x four variants; three are illustrative case studies, not a selection hierarchy

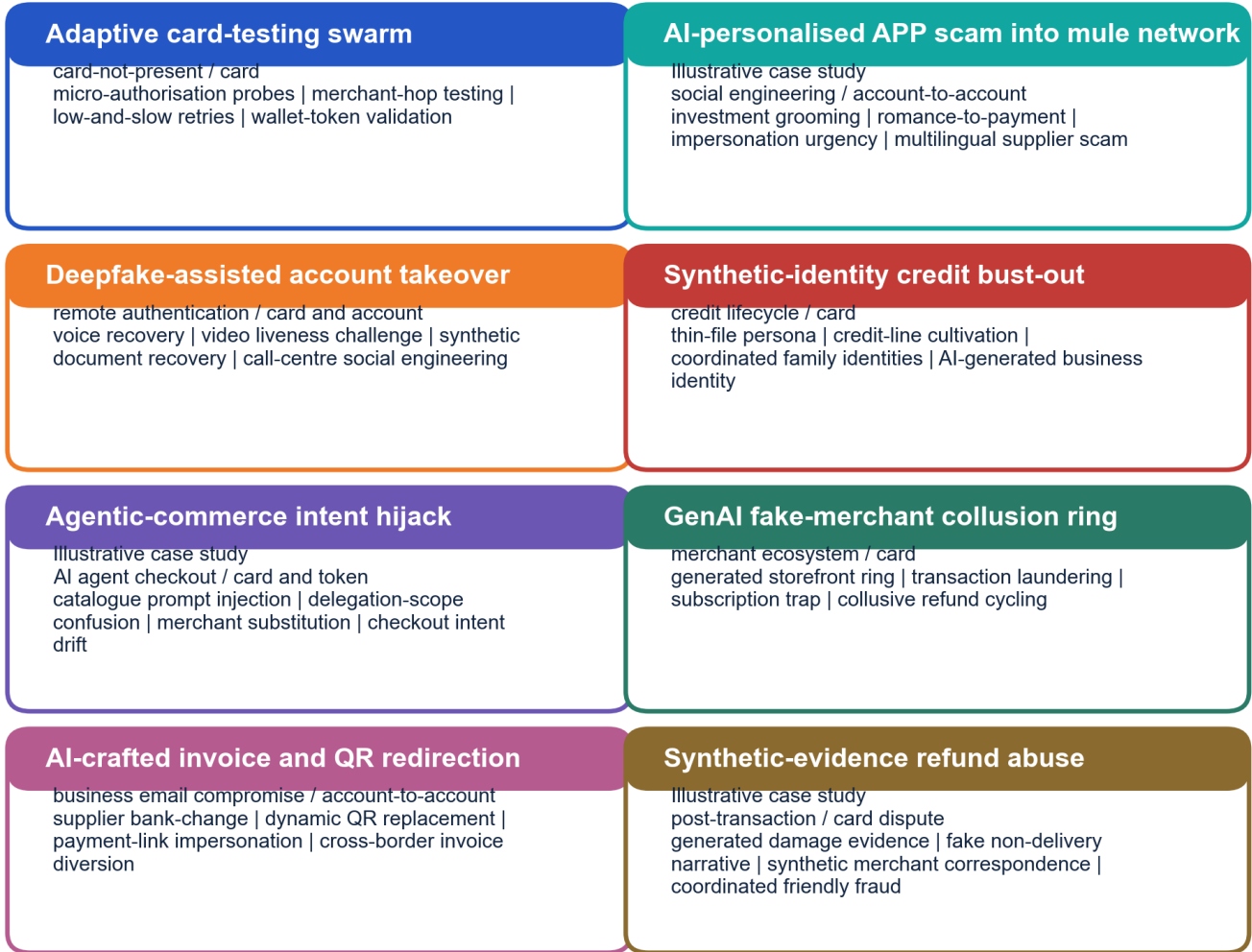


Figure 2. Attack atlas and taxonomy. All labels and variants are generated from attack_atlas.json [E6].

Three illustrative case studies show lifecycle coverage without creating a ranking.

Agentic-commerce intent hijack

A delegated shopping agent begins with legitimate user intent, encounters poisoned catalogue or merchant context, drifts outside its scope and completes checkout. The defense implication is verifiable intent and action traceability, not only transaction anomaly scoring. The approved scenario DSL defines four variants - catalogue prompt injection, delegation-scope confusion, merchant substitution, checkout intent drift - and the ordered sequence delegation, catalogue_poison, intent_drift, agent_checkout. [E6]

AI-personalised APP scam into a mule network

An LLM-assisted pretext sustains trust, introduces a new payee, tests a small transfer and escalates value. The event sequence lets the simulator expose beneficiary reuse and behavioral step-up signals across account-to-account payments. The approved scenario DSL defines four variants - investment grooming, romance-to-payment, impersonation urgency, multilingual supplier scam - and the ordered sequence trust_build, new_payee, small_test, escalated_transfer. [E6]

Synthetic-evidence refund abuse

A legitimate purchase and fulfilment are followed by forged damage or non-delivery evidence and a refund claim. The control surface shifts to evidence provenance, claim history and merchant fulfilment signals after authorization. The approved scenario DSL defines four variants - generated damage evidence, fake non-delivery narrative, synthetic merchant correspondence, coordinated friendly fraud - and the ordered sequence purchase, fulfilment, evidence_forge, refund_claim. [E6]

Safety boundary. These are synthetic research patterns. The system contains no cardholder data, credentials or live-payment connector; the attack atlas is compiled into bounded simulation parameters rather than exposed as a free-form attack service. [C1,E7]

Table 4. Attack-family roles

Family role	Count	Permitted use	Leakage control
Known	4	Supervised train / validation / test campaigns	Campaign-disjoint partitions
Adaptive search	2	Bandit search and hardening	Absent from initial supervised data
Final held-out	2	V0/V1/V2 generalization evaluation	Never enters search, training or calibration

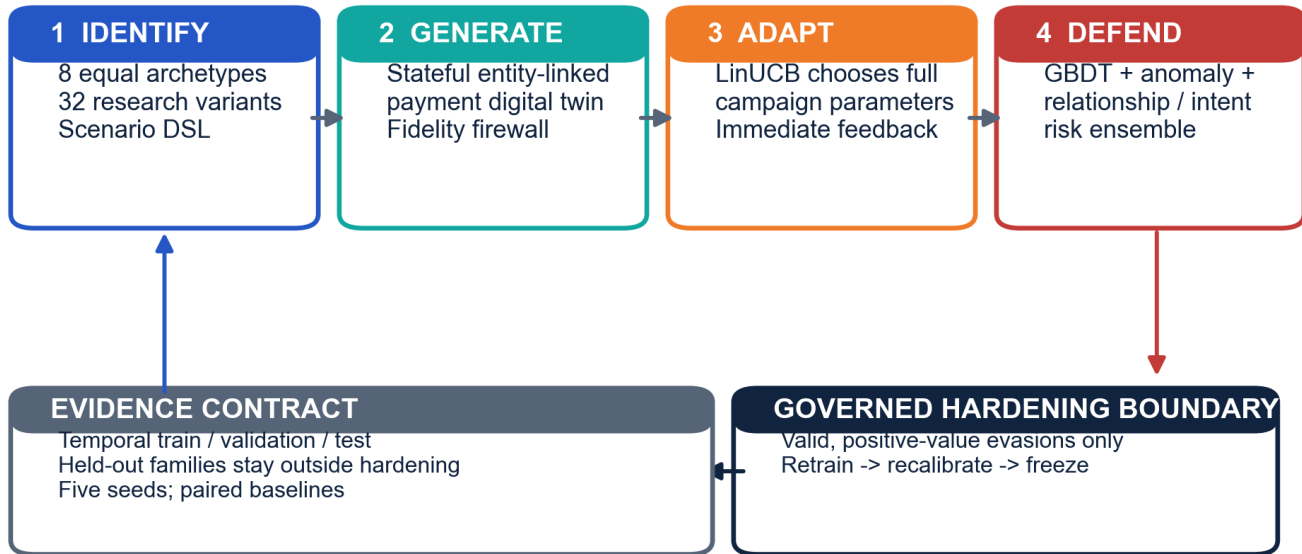
Source: architecture and benchmark contracts [E7,E8].

4 | System Architecture

AegisLoop is an offline closed loop with four separable responsibilities: hypothesis management, safe generation, adaptive search and governed defense hardening.

AegisLoop closes the attack-to-defense learning loop

Offline synthetic threat laboratory; no connection to live payment rails



Evidence source: docs/ARCHITECTURE.md and docs/BENCHMARKS.md

Figure 1. Closed-loop architecture. The diagram is generated from the documented system and evidence contracts [E7,E8].

The control boundary prevents the red team from silently changing the test.

Within a generation, the defender is frozen and only returns black-box outcomes. The attacker may update its LinUCB state, but cannot retrain the detector, change its threshold or admit invalid campaigns. Between generations, only fidelity-valid evasions with positive approved value are archived; the next defender is then trained, recalibrated on the unchanged legitimate validation split and frozen. [E7,E8]

The pipeline separates the object being tested from the evidence that validates it.

The simulator writes synthetic event features. The defender independently preprocesses and scores them. The fidelity firewall checks chronology, ranges, entity reuse and behavioral constraints, but cannot read detector probabilities or thresholds and cannot approve a reward. This separation matters: fidelity gates malformed scenarios; the defender supplies the adversarial outcome. [E7]

5 | Generate | Stateful Simulation and Fidelity

AegisLoop chooses controllable structural fidelity over untestable realism: campaigns are stateful and entity-linked by construction, while no claim is made that they reproduce a real portfolio distribution.

Simulator design

The hand-coded agent simulation creates customers, devices, merchants and beneficiaries, binds those entities to a scenario, and rolls out ordered events. Timestamps, velocity aggregates, reuse counts and relationship features evolve with the campaign. It is not CTGAN, TVAE, GaussianCopula, TabularARGN or another learned row-independent generator. [E7]

V2 fidelity diagnostic

The post-approval diagnostic rules out the trivial explanation that V2 received zero approved value because its campaigns failed validation. Every V2 campaign is valid; the table reports fidelity and action diversity directly from the 360-row artifact. [E5]

Seed	Valid campaigns	Mean fidelity	Minimum fidelity	Unique actions
20260812	72/72	0.927	0.837	33
20260813	72/72	0.918	0.838	32
20260814	72/72	0.919	0.824	37
20260815	72/72	0.927	0.837	31
20260816	72/72	0.924	0.831	37

Table 5. Campaign-diagnostic summary. Each seed contains 72 campaigns and 864 events [E5].

What the fidelity score does and does not mean

- It means the campaign passed schema, range, chronology, behavioral and entity-reuse checks defined by the simulator. [E7]
- It helps prevent a policy from earning reward with malformed or structurally incoherent synthetic traffic. [E7]
- It does not mean the campaign matches a real issuer's marginal distribution, temporal signature or fraud loss rate; those require authorized reference data and external validation. [E7,R3]

6 | Adapt | Contextual-Bandit Red Team

The attacker is genuine lightweight RL: a contextual bandit learns which complete campaign configuration to try next from immediate black-box defender feedback.

Why a contextual bandit, not a full MDP

Each decision chooses one complete 12-event campaign from a finite action pool and receives an immediate scalar reward after the frozen defender scores it. There is no need to estimate delayed long-horizon value; LinUCB supplies transparent exploration and exploitation with a small data budget. The state summarizes recent detector outcomes; the action encodes attack family and bounded intensity, timing, reuse and evasion parameters. [E7,E8]

Reward is normalized, decomposed and falsifiable.

$$R = F \times (V / 500 + 2.5 \times (1 - D) - 0.16 \times D - 2.5 \times N / 1000 + 0.20 \times I_{\text{novel}})$$

F is fidelity in [0,1], V is approved synthetic value, D is event detection rate, N is campaign event count and I_{novel} marks the first successful discretized pattern. Invalid campaigns receive -1 and zero component terms. The coefficients are fixed engineering design priors, not weights fitted to the five reported seeds: V/500 puts hundreds of approved-value units on the evasion scale; 2.5 makes full evasion the dominant objective; -0.16D penalizes detection; -2.5N/1000 charges 0.0025 per event to discourage gratuitous length; and the 0.20 novelty bonus is deliberately subordinate to evasion. Every term, pre-fidelity reward and final reward is logged per campaign. [E7-E9]

The novelty component is instrumented, not credited with performance.

Full five-seed ablation	Mean reward	SD	Bootstrap 95% CI
Novelty on	0.807	0.457	[0.519, 1.212]
Novelty zeroed	0.824	0.349	[0.574, 1.105]
Paired on - off	-0.017	0.159	[-0.137, 0.121]

Table 2. Reward ablation. Two-sided paired Wilcoxon $p=0.625$; novelty is not a performance driver [E1,E2].

7 | Defend | Ensemble and Hardening

The blue team combines complementary signals, then recalibrates each hardened generation on the same disjoint validation window before any test metric is computed.

Layer	Role	Why it matters in this threat model
Supervised gradient boosting	Learn labeled tabular interactions	Strong known-pattern baseline and calibrated ranking
Isolation scoring	Surface distributional outliers	Covers unusual behavior without an attack label
Relationship / intent risk	Encode entity reuse and delegated-intent inconsistency	Captures campaign structure beyond a single row
Policy action	Approve, step-up, hold or decline	Maps score to an operational response vocabulary

Table 6. Defender ensemble design [E7].

Hardening uses only evidence that an operational control plane could govern.

A successful red-team campaign must be fidelity-valid, produce positive approved synthetic value and evade at least part of the frozen defender. Only then can it enter the next defender's training data. Validation legitimate traffic remains unchanged, each generation computes its own threshold and test traffic remains untouched. This creates two real transitions, V0->V1 and V1->V2, without test-set feedback. [E3,E7,E8]

The design is compatible with - but separate from - live decisioning.

In a production architecture, AegisLoop outputs scenario definitions, validated synthetic campaigns, feature-gap findings and candidate challenger data. A network or issuer decision engine remains the system of record. AegisLoop does not sit in the authorization path and makes no live approve/decline decision. [E7]

8 | Experimental Methodology

The evidence protocol is paired, seed-robust and leakage-controlled; its remaining weakness is statistical scale, not hidden test reuse.

Table 3. Temporal split and calibration contract

Split	Legitimate rows	Temporal window	Purpose
Train	16,800	0%-60%	Fit preprocessor, supervised model and anomaly model
Validation	5,600	60%-80%	Select each defender's threshold at 1% target FPR
Test	5,600	80%-100%	Report FPR, recall, precision and ROC-AUC only

The audit records zero validation/test event-ID overlap, zero known-campaign overlap across partitions, separately generated held-out campaigns and test_rows_used_for_threshold_selection=false. [E3]

Equal-budget red-team comparison

Random search, rule mutation and LinUCB receive the same frozen V0, search families, validity gate, action-pool size and 72 campaign evaluations of 12 events each. The five fixed seeds are 20260812-20260816. Quick-mode outputs exist only as a liveness check; all paper results use the full protocol. [E1,E2,E8]

Statistical plan

- Report sample mean and SD across five seeds, plus a nonparametric bootstrap 95% CI for the seed-level mean using 20,000 draws. [E1]
- Use one-sided paired Wilcoxon signed-rank tests for pre-specified directional comparisons of bandit reward against each baseline and later defender generations against earlier ones. [E1]
- Use a two-sided paired Wilcoxon test for the novelty on/off ablation because either direction is substantively possible. [E1]
- With five non-zero pairs, the smallest exact one-sided p-value is 0.03125; p-values therefore accompany raw seed points and interval estimates. [E1,E2,E8]

Prevalence protocol

The seed-20260812 held-out slice contains 580 attack events and 5,600 legitimate events (9.39% attack prevalence). The exact deployment stress scenario importance-reweights the same scores to 0.015%, matching the ECB/EBA reported H1 2023 share of fraudulent card payments by transaction count. A 400-repeat literal downsampling diagnostic is retained but contains one positive per repeat and is treated as a high-variance sensitivity check. [E4,E8,R5]

9 | Results

The adaptive attacker is consistently stronger than equal-budget baselines, and defender efficacy improves through V2; the result is credible because its nulls and boundary failures are visible beside its wins.

Outcome	Estimate across five seeds	Significance / interpretation
Bandit reward	0.807 [0.519, 1.212]	Wins 5/5 vs random; $p=0.03125$
Random reward	0.126 [0.020, 0.235]	Paired mean gap: 0.681 [0.489, 0.984]
Rule-mutation reward	0.503 [0.167, 1.047]	Bandit wins 5/5; $p=0.03125$
V0 held-out ROC-AUC	0.9806 [0.9755, 0.9853]	Validation-calibrated threshold
V2 held-out ROC-AUC	0.99993 [0.99984, 1.00000]	Final families never entered hardening
V0 -> V1 fresh value reduction	92.1% [85.9%, 98.1%]	Reward improves on 5/5; $p=0.03125$
V1 -> V2 fresh reward	Mean reduction 0.132 [0.033, 0.239]	4 wins, 1 tie; $p=0.0625$, directional

Table 1. Headline results. Brackets are bootstrap 95% CIs for seed-level means or paired differences [E1,E2].

The contextual bandit wins every paired seed against both baselines

Diamonds: mean; bars: bootstrap 95% CI; circles: all five fixed seeds

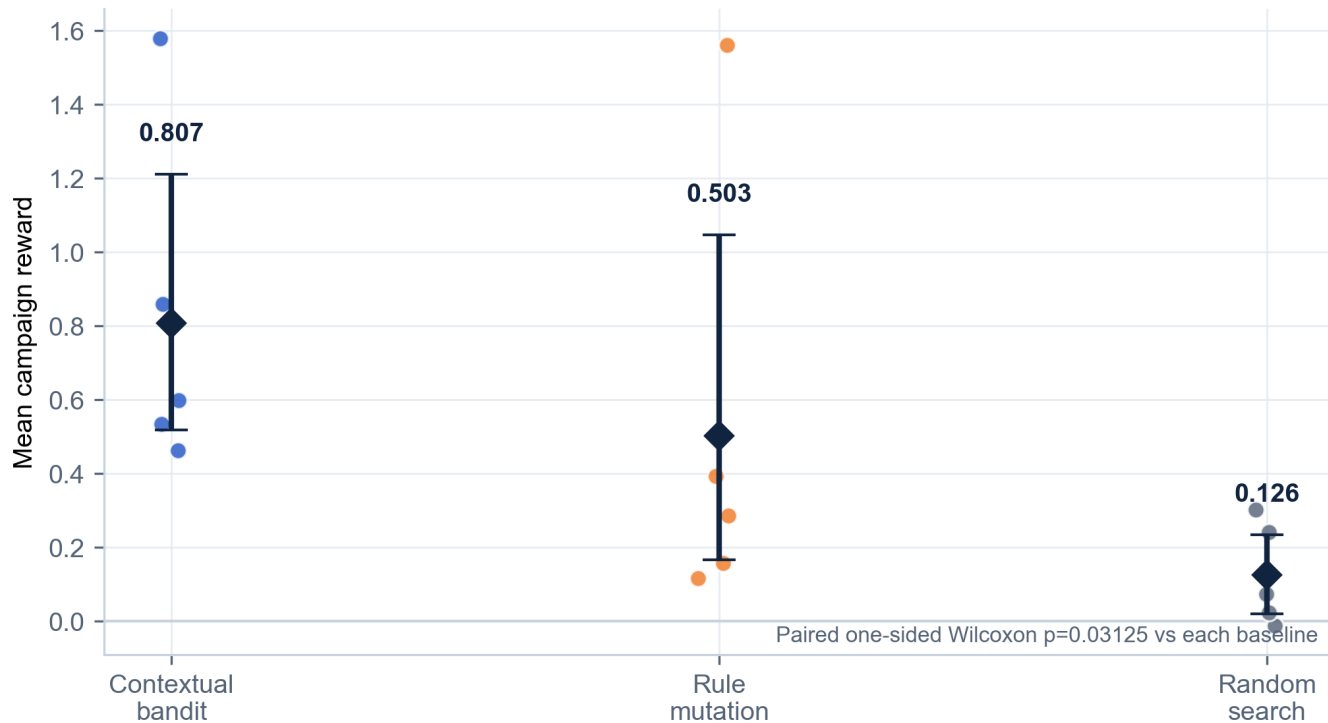


Figure 3. Equal-budget reward comparison. All five fixed seeds are plotted; no single seed is promoted as the result [E1,E2].

Held-out-family efficacy improves without forcing test FPRs to match.

Held-out-family performance improves through two hardening cycles

Lines: five-seed mean; bands: bootstrap 95% CI

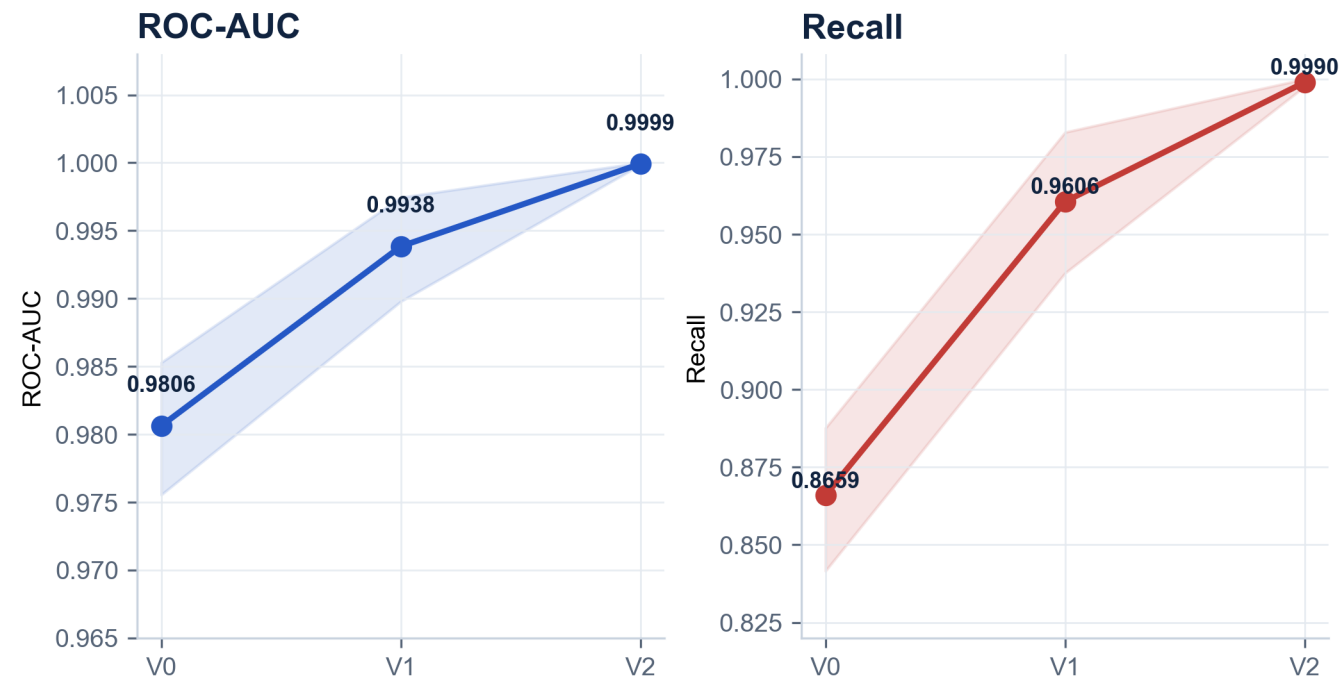


Figure 4. Held-out ROC-AUC and recall progression. Lines are five-seed means and bands are bootstrap 95% CIs [E1].

Metric	V0 mean +/- SD	V1 mean +/- SD	V2 mean +/- SD
ROC-AUC	0.9806 +/- 0.0062	0.9938 +/- 0.0049	0.99993 +/- 0.00010
Recall	86.59% +/- 2.91%	96.06% +/- 2.92%	99.90% +/- 0.15%
Precision at experiment prevalence	90.68% +/- 1.92%	90.73% +/- 1.06%	91.28% +/- 1.72%
Legitimate test FPR	0.911% +/- 0.163%	1.007% +/- 0.111%	0.982% +/- 0.200%

Table 7. Held-out-family results. Each threshold is calibrated independently on validation at a 1% target FPR; test FPRs are natural realizations [E1,E3].

The V2 zero is meaningful only inside the matched-pool diagnostic.

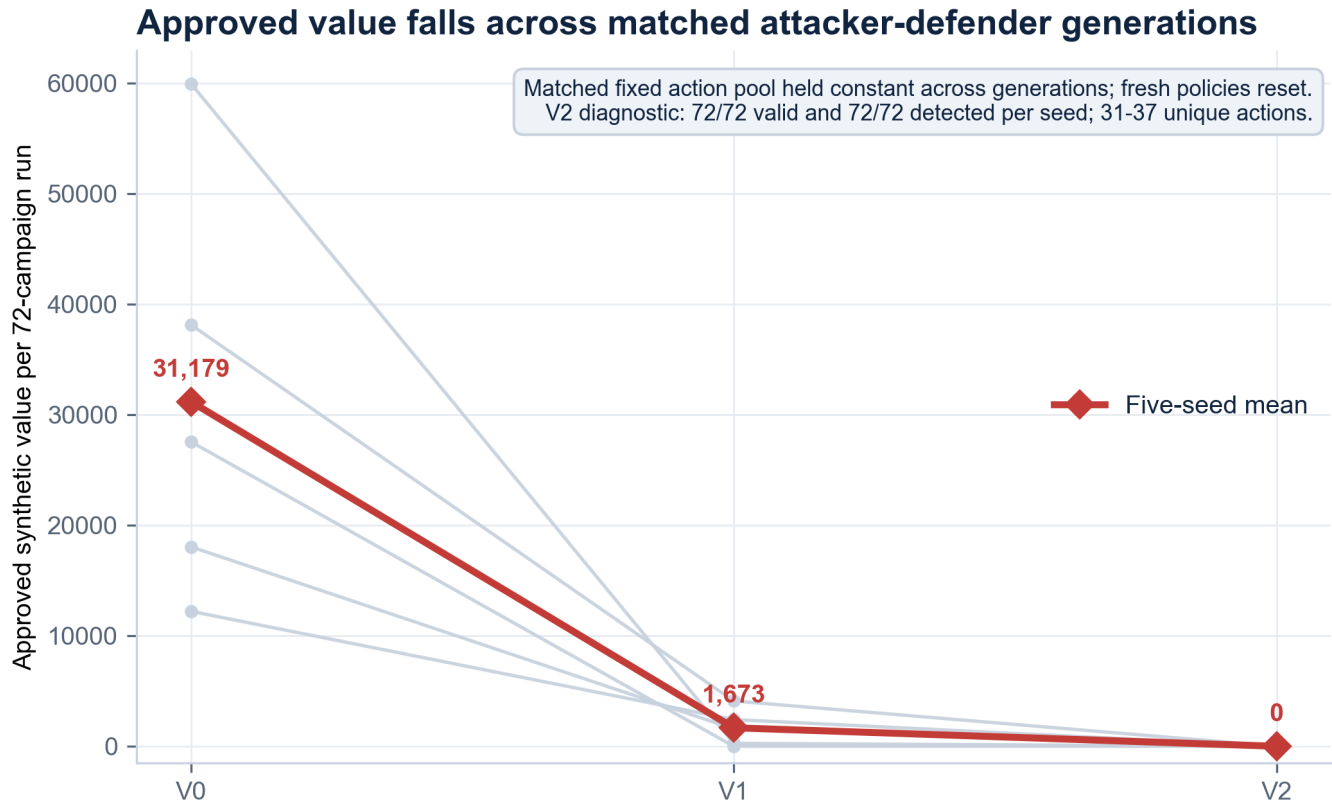


Figure 5. Approved synthetic value across defender generations. The action pool and random stream are matched across generations; the caveat is embedded in the chart [E1,E2,E5].

Against a matched, fixed action pool held constant across generations, the hardened defender (V2) detected every campaign across five seeds. The underlying artifact contains 360/360 valid campaigns, 360/360 fully detected campaigns and zero approved value; each seed selects 31-37 unique actions. This is not evidence about an independently resampled V2 action space, unbounded adversaries or live traffic. [E5]

The V1->V2 reward comparison is lower on four seeds and tied on one, with one-sided paired Wilcoxon $p=0.0625$. The evidence therefore shows no observed oscillation through one additional generation pair; it does not eliminate the instability risk identified in the Adyen study. [E1,E2,R2]

The novelty ablation is null, and that improves the specification.

Novelty-on minus novelty-off reward is -0.017 (95% CI -0.137 to 0.121; two-sided paired Wilcoxon $p=0.625$). The novelty term remains logged for research traceability but is not presented as a driver of attacker quality. The result concentrates the product claim on approved value, evasion and controlled search. [E1,E2]

The complete numerical ablation is reported once in Table 2 (Section 6); the interpretation here is unchanged and is intentionally shown alongside the other efficacy and boundary findings. [E1,E2]

Prevalence is the strongest constraint on operational precision.

High lab precision does not survive realistic prevalence

Seed 20260812; exact importance reweighting to 0.015%

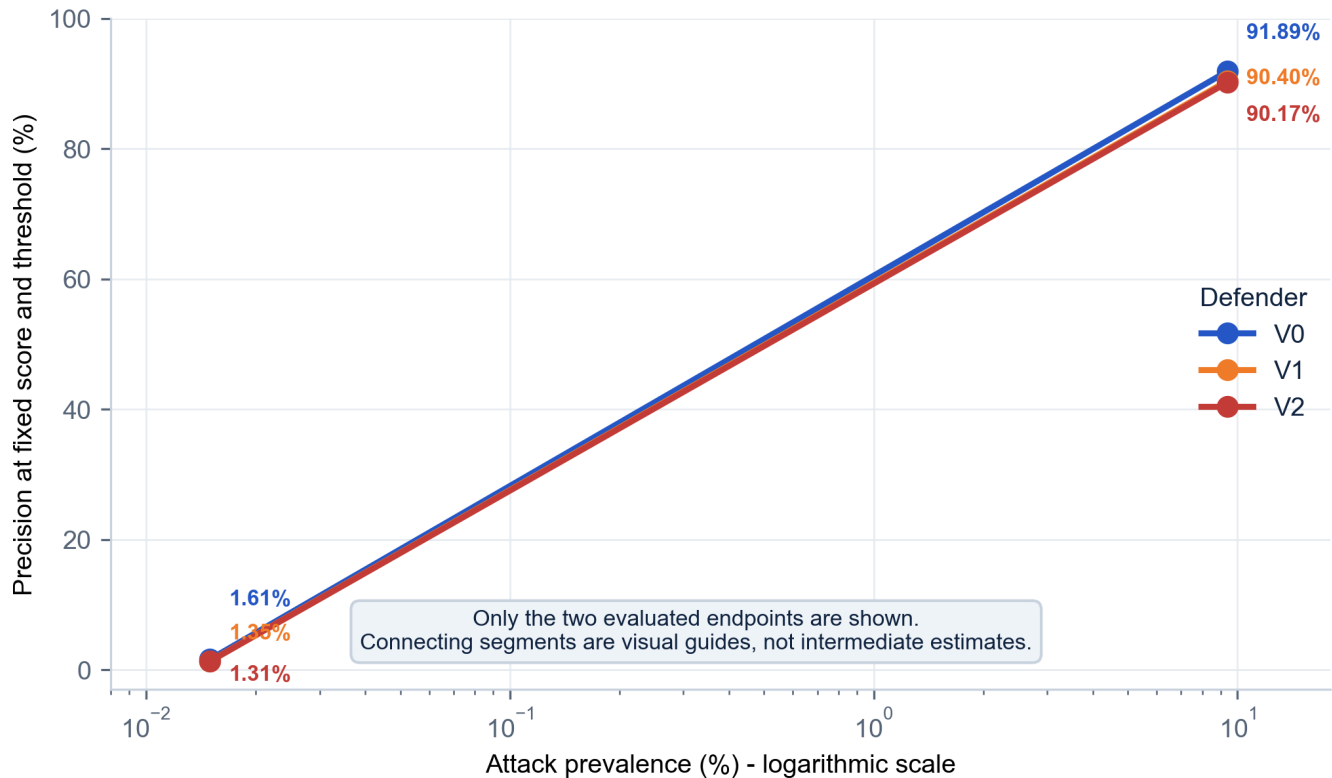


Figure 6. Precision versus prevalence. Only the two artifact-backed endpoints are plotted; connecting segments are visual guides. Seed 20260812; full range in Appendix Table A3 [E4].

For V2, precision falls from 90.17% at 9.39% experimental prevalence to 1.31% at exactly reweighted 0.015% prevalence (seed 20260812; full range in Appendix Table A3). AUC and FPR remain fixed for the same scores and threshold; precision changes because false positives vastly outnumber true positives when fraud is rare. The lab precision must not be presented as a production estimate. [E4]

10 | Real-World Feasibility and Business Impact

The feasible product is an offline adversarial assurance layer that feeds governed challenger development - not an autonomous attacker and not a replacement for network decisioning.

Stage	Production interface	Control / evidence
1. Observe	Approved outcome labels, feature contracts and drift summaries	No raw cardholder data enters the synthetic lab by default
2. Model	Parameterize the digital twin from authorized aggregates	Versioned scenarios; lineage; fidelity tests
3. Challenge	Attack frozen champion/challenger snapshots offline	Equal budgets; validity firewall; reproducible seeds
4. Harden	Export valid evasions and feature-gap findings	Human approval; validation recalibration; model-risk review
5. Release	Shadow or champion/challenger deployment	Kill switch; drift and false-positive monitoring

Table 8. Directional deployment path. This is an architectural proposal, not measured implementation performance [E7].

Integration path

AegisLoop can sit upstream of an issuer's model-development environment. It consumes governed schemas, calibrated aggregate constraints and frozen model APIs; it returns scenario DSL versions, campaign diagnostics, evasive examples and proposed feature gaps. The online score remains owned by existing authorization infrastructure. [E7]

Scalability

Campaign evaluation is embarrassingly parallel across action candidates and seeds, while policy updates are lightweight linear algebra. The scenario DSL isolates research expansion from detector code. A credible scale-up plan is therefore to distribute simulator rollouts, retain deterministic seeds and promote only versioned aggregates across governance boundaries. No latency or throughput number is claimed because this prototype has not been benchmarked as production infrastructure.

Commercial value levers

The business case has four measurable levers: fraud loss avoided through earlier discovery, false-decline cost avoided through better calibrated controls, analyst effort reduced through prioritized evasions, and model-release risk reduced through repeatable adversarial tests. The evidence supports the mechanism, not an ROI estimate; portfolio loss rates, review costs and authorization economics must be supplied by the deployment partner before monetization is quantified.

Alignment with Mastercard's public direction

Decision Intelligence Pro publicly emphasizes relationship-aware, contextual and real-time risk assessment. AegisLoop is complementary: it creates governed synthetic pressure tests for the entity and intent signals that a decision engine may consume. This is a directional comparison only; no equivalence to Mastercard's technology or published performance is asserted. [R4]

11 | Limitations and Responsible Use

AegisLoop has earned a strong laboratory result, not a production performance claim; the remaining gaps are explicit enough to form the next validation agenda.

Synthetic-only evidence

The simulator has no authorized real reference corpus, so structural validity cannot establish statistical or behavioral equivalence to a live portfolio. External validation must compare entity-level temporal and graph signatures. [E7,R3]

Matched-pool V2 scope

The V2 claim holds against a matched, fixed action pool shared across generations. It does not establish resistance to an independently resampled, larger or open-ended attack space. [E5]

Small seed count

Five paired seeds allow an exact one-sided $p=0.03125$ only when every difference has the expected sign. Confidence intervals remain wide for attacker reward. [E1,E2,E8]

Limited stability horizon

Only $V0 \rightarrow V1 \rightarrow V2$ is observed. The $V1 \rightarrow V2$ reward result is directional at $p=0.0625$; additional generations are required to test oscillation. [E1,R2]

Prevalence transfer

Experiment prevalence is 9.39% in the seed-20260812 held-out slice. Exact reweighting shows precision near 1.3%-1.6% at 0.015%, so production thresholds and review capacity require portfolio calibration. [E4]

No live-operational proof

The prototype has no live endpoints, no measured production latency, no model-risk approval and no measured ROI. Deployment claims are architectural and qualitative. [E7]

Responsible-use controls

- Keep the red-team policy inside a synthetic, access-controlled environment with no live payment-rail connector.
- Use only synthetic, anonymized or explicitly authorized inputs consistent with the competition rules. [C1]
- Require human approval before an evasion enters training or a challenger model reaches shadow deployment.
- Publish negative results and provenance alongside performance summaries so presentation cannot outrun evidence.

12 | Conclusion

AegisLoop's defensible novelty is a campaign-level extension of single-transaction RL fraud-evasion attacks that closes the loop into adversarial defender hardening.

The solution converts the challenge's Identify-Generate-Defend brief into one governed feedback system. Eight equal-priority archetypes define the search space; a stateful entity-linked simulator generates campaigns; a contextual bandit finds blind spots more effectively than equal-budget random and rule baselines; and a leakage-controlled ensemble improves across two hardening cycles. [C1,E1-E8]

The evidence is strongest where it is most bounded. The five-seed bandit advantage is paired and statistically positive. Held-out AUC and recall improve through V2. The V2 matched-pool diagnostic is non-trivial because every campaign is valid and action use remains diverse. At the same time, the novelty bonus does not help, later-generation stability is not conclusive, and realistic prevalence collapses precision. Those are not presentation defects; they are the difference between an innovation demo and an auditable model-development instrument. [E1-E5]

The next decision is therefore concrete: validate the digital twin against authorized aggregate behavioral signatures, expand the independently sampled action space and generation horizon, then evaluate the resulting challenger in shadow mode with portfolio-specific calibration. Until that evidence exists, AegisLoop should be positioned exactly as built - an offline adaptive fraud-defense laboratory.

References

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- [R4] Mastercard. *Mastercard supercharges consumer protection with gen AI*, 1 February 2024. [Mastercard Newsroom](#).
- [R5] European Central Bank and European Banking Authority. *ECB and EBA publish joint report on payment fraud*, 1 August 2024. [ECB press release](#).
- [C1] Mastercard Innovation Challenge 2026. *Competition Rules Overview*. Private competition page copied to Comp_Rules_Overview.txt; accessed in the authorized local repository.

Appendix A | Full Statistical Tables

Table A1. Raw seed-level primary outcomes

Seed	Bandit reward	Rule reward	Random reward	V0 AUC	V1 AUC	V2 AUC	V0 value	V1 value	V2 value
20260812	1.579	1.562	0.303	0.9886	0.9961	0.99979	59,952	267	0
20260813	0.859	0.286	0.242	0.9724	0.9867	1.00000	38,133	4,100	0
20260814	0.463	0.157	-0.011	0.9772	0.9928	0.99984	27,549	0	0
20260815	0.535	0.394	0.074	0.9815	1.0000	1.00000	12,214	2,417	0
20260816	0.599	0.117	0.024	0.9834	0.9936	1.00000	18,048	1,579	0

Source: seed_results.csv [E2]. 'Value' is fresh-attacker approved synthetic value.

Table A2. Calibration audit

Defender	Source split	Rows	Target FPR	Validation FPR	Threshold
V0	legitimate_validation_temporal_60_80pct	5,600	1.00%	1.00%	0.138344
V1	legitimate_validation_temporal_60_80pct	5,600	1.00%	1.00%	0.136563
V2	legitimate_validation_temporal_60_80pct	5,600	1.00%	1.00%	0.138693

Validation/test event-ID overlap=0; test rows used for threshold selection=false [E3].

Table A3. Seed-20260812 prevalence metrics

Defender	Scenario	Prevalence	Fraud rows	Repeats	ROC-AUC	Precision	Recall	FPR
V0	Experiment	9.3851%	580	1	0.9886	91.89%	89.83%	0.821%
V0	Literal downsample	0.0179%	1	400	0.9843	1.91%	89.75%	0.821%
V0	Exact reweight	0.0150%	580	1	0.9886	1.61%	89.83%	0.821%
V1	Experiment	9.3851%	580	1	0.9961	90.40%	97.41%	1.071%
V1	Literal downsample	0.0179%	1	400	0.9931	1.59%	97.25%	1.071%
V1	Exact reweight	0.0150%	580	1	0.9961	1.35%	97.41%	1.071%
V2	Experiment	9.3851%	580	1	0.9998	90.17%	99.66%	1.125%
V2	Literal downsample	0.0179%	1	400	0.9997	1.55%	99.50%	1.125%
V2	Exact reweight	0.0150%	580	1	0.9998	1.31%	99.66%	1.125%

Source: prevalence_metrics.csv [E4]. The exact-reweight rows support the standalone precision claims. Literal downsampling uses one fraud row per repeat at this test-set size and is a high-variance sensitivity check.

Table A4. V2 campaign diagnostics by seed

Seed	Campaigns	Valid	Fully detected	Events	Approved value	Mean fidelity	Min fidelity	Unique actions	Families
20260812	72	72	72	864	0.0	0.9266	0.8369	33	6
20260813	72	72	72	864	0.0	0.9178	0.8376	32	6
20260814	72	72	72	864	0.0	0.9193	0.8238	37	6
20260815	72	72	72	864	0.0	0.9269	0.8371	31	6
20260816	72	72	72	864	0.0	0.9244	0.8307	37	6

Source: v2_campaign_diagnostics.csv, artifact version 1.0 [E5].

Appendix B | Reproducibility and Evidence Ledger

Reproduce the evidence

<code>.venv\Scripts\python -m pytest</code>
<code>.venv\Scripts\python -m ruff check src experiments tests app.py scripts</code>
<code>.venv\Scripts\python experiments\run_multiseed.py --seeds 20260812,20260813,20260814,20260815,20260816</code>
<code>.venv\Scripts\python scripts\make_figures.py</code>
<code>.venv\Scripts\python scripts\build_paper.py</code>

The full multi-seed command writes both `seed_results.csv` / `aggregate.json` and the versioned V2 campaign diagnostic. The figure script reads approved artifacts and emits PNG/SVG files. The paper builder rereads the same artifacts; no result is copied from an earlier withdrawn headline. [E1-E5,E8]

Approved evidence ledger

ID	Path	What it supports
E1	artifacts/full_multiseed/aggregate.json	Five-seed means, SDs, CIs and significance tests
E2	artifacts/full_multiseed/seed_results.csv	Raw per-seed outcomes
E3	artifacts/precomputed/calibration_audit.json	Temporal split, threshold source and leakage audit
E4	artifacts/precomputed/prevalence_metrics.csv	Seed-20260812 prevalence scenarios and metrics
E5	artifacts/precomputed/v2_campaign_diagnostics.csv	Raw V2 campaign validity, detection, fidelity and action diversity
E6	artifacts/precomputed/attack_atlas.json	Eight archetypes, 32 variants and scenario metadata
E7	docs/ARCHITECTURE.md	System, simulator, reward and deployment architecture
E8	docs/BENCHMARKS.md	Protocol contract, statistics and full-vs-quick distinction
E9	RESULTS.md	Phase 1 verified narrative and result tables
E10	PHASE1_CHANGELOG.md	Finding-to-remediation changelog

Repository URL: <https://github.com/Hermes-25/AegisLoop>

Working prototype URL: <https://aegisloop-mcic-2026.vercel.app>

Document status: evidence-locked solution walkthrough. Quantitative claims are derived from the approved artifacts above; external grounding is limited to references [R1]-[R5] and the competition rules [C1].